2.2 Logical Schema Mapping

The transformation of the Conceptual ER/EER model into a Relational Schema adheres to formal relational mapping algorithms (mapping strong entities, weak entities, binary 1:1, 1:N, recursive 1:1, and specialization hierarchies).

Specialization Mapping Strategy

For the DOCTOR hierarchy with **Total and Disjoint** specialization, the **Separate Relations for Superclass and Subclasses (Option 8A / Joined Table Strategy)** is employed:

- A base table DOCTOR retains all common attributes (DoctorID, UserID, FullName, Phone, Email, LicenseNumber).
- Specific tables GENERAL_PRACTITIONER and SPECIALIST store subtype-specific relationships (SpecialtyID) and share the primary key DoctorID, functioning simultaneously as a **Foreign Key** referencing DOCTOR(DoctorID).
- This approach eliminates NULL markers, prevents schema update anomalies, and enforces referential integrity for specialist-to-specialty foreign keys.

Relational Schema Definition

Notation: **PK (Bold and Underlined)** indicates Primary Key; **FK (Italic with Asterisk)** indicates Foreign Key.

1. **USER_ACCOUNT** (**UserID**, Username, PasswordHash, Role, AccountStatus, CreatedAt)
2. **PATIENT** (**PatientID**, **UserID**, FullName, DateOfBirth, Gender, Phone, Email, Address, BloodType, Allergies, ChronicDiseases, EmergencyContactName, EmergencyContactPhone)
    - *FK:* UserID → USER_ACCOUNT(UserID)

3. **SPECIALTY** (**SpecialtyID**, SpecialtyName, Description)
4. **DOCTOR** (**DoctorID**, **UserID**, FullName, Phone, Email, LicenseNumber)
    - *FK:* UserID → USER_ACCOUNT(UserID)

5. **GENERAL_PRACTITIONER** (****DoctorID****)
    - *PK/FK:* DoctorID → DOCTOR(DoctorID)

6. **SPECIALIST** (****DoctorID****, **SpecialtyID**)
    - *PK/FK:* DoctorID → DOCTOR(DoctorID)

    - *FK:* SpecialtyID → SPECIALTY(SpecialtyID)

7. **DOCTOR_SCHEDULE** (**ScheduleID**, **DoctorID**, ScheduleDate, StartTime, EndTime, AvailabilityStatus)
    - *FK:* DoctorID → DOCTOR(DoctorID)

8. **APPOINTMENT** (**AppointmentID**, **PatientID**, **DoctorID**, **ScheduleID**, **BookedByUserID**, **FollowUpFromApptID**, AppointmentDateTime, EstimatedDurationMinutes, AppointmentType, Status, Reason)

- *FK:* PatientID → PATIENT(PatientID)
- *FK:* DoctorID → DOCTOR(DoctorID)
- *FK:* ScheduleID → DOCTOR_SCHEDULE(ScheduleID)
- *FK:* BookedByUserID → USER_ACCOUNT(UserID)
- *FK:* FollowUpFromApptID → APPOINTMENT(AppointmentID) (Recursive). FollowUpFromApptID is NULL if the appointment is not a follow-up. If non-NULL, it must reference an existing valid preceding APPOINTMENT(AppointmentID).

9. **CONSULTATION_SESSION** (**SessionID**, **AppointmentID**, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingURL, DiagnosisNotes)
   - *FK:* AppointmentID → APPOINTMENT(AppointmentID) (NOT NULL, UNIQUE)
10. **MEDICAL_HISTORY** (**MedicalHistoryID**, **SessionID**, RecordDate, Diagnosis, Symptoms, ProgressNotes)
    - *FK:* SessionID → CONSULTATION_SESSION(SessionID)
11. **MEDICATION** (**MedicationID**, MedicationName, Description)
12. **PRESCRIPTION** (**PrescriptionID**, **SessionID**, DiagnosisICD, PrescriptionDate, Instructions)
    - *FK:* SessionID → CONSULTATION_SESSION(SessionID)
13. **PRESCRIPTION_ITEM** (**PrescriptionItemID**, **PrescriptionID**, **MedicationID**, Dosage, Frequency, Duration, SpecialInstructions)
    - *FK:* PrescriptionID → PRESCRIPTION(PrescriptionID)
    - *FK:* MedicationID → MEDICATION(MedicationID)

2.3 Normalization Verification (Formal Proofs: 1NF → 3NF / BCNF)

Foundational Definitions

- **1NF (First Normal Form):** Every attribute domain contains only atomic (indivisible) values; relations contain no repeating groups or nested relations.
- **2NF (Second Normal Form):** The relation is in 1NF and every non-prime attribute is fully functionally dependent on every candidate key (no partial key dependencies: $\forall X \rightarrow A$, where X is a strict subset of any candidate key).
- **3NF (Third Normal Form):** The relation is in 2NF and for every non-trivial functional dependency $X \rightarrow A$, either X is a superkey or A is a prime attribute (no transitive dependencies of non-prime attributes on candidate keys).
- **BCNF (Boyce-Codd Normal Form):** The relation is in 1NF and for every non-trivial functional dependency $X \rightarrow A$, X must strictly be a superkey.

Proof for Relation: USER_ACCOUNT

- **Attribute Set: R1 = {UserID, Username, PasswordHash, Role, AccountStatus, CreatedAt}**
- **Functional Dependencies (FDs):**
    - FD1.1: UserID → {Username, PasswordHash, Role, AccountStatus, CreatedAt}
    - FD1.2: Username → {UserID, PasswordHash, Role, AccountStatus, CreatedAt}
- **Candidate Keys (CKs):** CK1 = {UserID}, CK2 = {Username}
- **Prime Attributes:** {UserID, Username}
- **Non-Prime Attributes:** {PasswordHash, Role, AccountStatus, CreatedAt}

Verification Steps:

1. **1NF:** Every attribute holds atomic scalar values. No composite or multi-valued fields exist. ⇒ **1NF holds.**
2. **2NF:** The candidate keys CK1 and CK2 are singletons (cardinality = 1). No proper subset of any candidate key can exist. Thus, partial dependency is mathematically impossible. ⇒ **2NF holds.**
3. **3NF:** For FD1.1, X = {UserID} is a superkey. For FD1.2, X = {Username} is a superkey. There are no transitive dependencies to non-prime attributes. ⇒ **3NF holds.**
4. **BCNF:** For all non-trivial dependencies X → Y, the determinant X is a valid superkey. ⇒ **BCNF holds.**

Proof for Relation: PATIENT

- **Attribute Set:** R2 = {PatientID, UserID, FullName, DateOfBirth, Gender, Phone, Email, Address, BloodType, Allergies, ChronicDiseases, EmergencyContactName, EmergencyContactPhone}
- **Functional Dependencies (FDs):**
    - FD2.1: PatientID → {UserID, FullName, DateOfBirth, Gender, Phone, Email, Address, BloodType, Allergies, ChronicDiseases, EmergencyContactName, EmergencyContactPhone}
    - FD2.2: UserID → {PatientID, FullName, DateOfBirth, Gender, Phone, Email, Address, BloodType, Allergies, ChronicDiseases, EmergencyContactName, EmergencyContactPhone}
- **Candidate Keys:** CK1 = {PatientID}, CK2 = {UserID}

- **Prime Attributes:** {PatientID, UserID}

- **Non-Prime Attributes:** All remaining attributes.

Verification Steps:

1. **1NF:** Each attribute holds atomic scalar types. Baseline values (BloodType, Allergies, ChronicDiseases) represent flat baseline clinical notations, ensuring atomicity. ⇒ **1NF holds.**
2. **2NF:** Candidate keys are singleton sets. No non-prime attribute can partially depend on a key subset. ⇒ **2NF holds.**
3. **3NF:** For both FD2.1 and FD2.2, the left-hand sides are superkeys. There are no non-key dependencies (e.g., Phone ↛ Address). ⇒ **3NF holds.**
4. **BCNF:** Every determinant is a candidate key/superkey. ⇒ **BCNF holds.**

Proof for Relation: DOCTOR

- **Attribute Set:** R3 = {DoctorID, UserID, FullName, Phone, Email, LicenseNumber}

- **Functional Dependencies (FDs):**
    - FD3.1: DoctorID → {UserID, FullName, Phone, Email, LicenseNumber}

    - FD3.2: LicenseNumber → {DoctorID, UserID, FullName, Phone, Email}

    - FD3.3: UserID → {DoctorID, FullName, Phone, Email, LicenseNumber}

- **Candidate Keys:** CK1 = {DoctorID}, CK2 = {LicenseNumber}, CK3 = {UserID}

Verification Steps:

1. **1NF:** All attribute values are strictly atomic. ⇒ **1NF holds.**
2. **2NF:** All candidate keys contain exactly one attribute. No partial dependencies exist. ⇒ **2NF holds.**
3. **3NF & BCNF:** Determinants {DoctorID}, {LicenseNumber}, and {UserID} are all candidate keys (superkeys). Subtype-specific fields have been segregated into subtype relations, eliminating null-induced and transitive anomalies. ⇒ **BCNF holds.**

Proof for Relations: GENERAL_PRACTITIONER and SPECIALIST

- **GENERAL_PRACTITIONER:**
    - R4A = {DoctorID}, CK = {DoctorID}. Contains only the prime key attribute. Trivially in **BCNF**.
- **SPECIALIST:**
    - R4B = {DoctorID, SpecialtyID}, FD: DoctorID → SpecialtyID.
    - CK = {DoctorID}. Since DoctorID is a superkey, and each specialist belongs to exactly one specialty (BR18), the relation satisfies **BCNF**.

Proof for Relation: SPECIALTY

- **Attribute Set:** R5 = {SpecialtyID, SpecialtyName, Description}

- **Functional Dependencies:**
    - FD5.1: SpecialtyID → {SpecialtyName, Description}

    - FD5.2: SpecialtyName → {SpecialtyID, Description}

- **Candidate Keys:** CK1 = {SpecialtyID}, CK2 = {SpecialtyName}

- **Verification:** Both determinants are candidate keys. ⇒ **BCNF holds.**

Proof for Relation: DOCTOR_SCHEDULE

- **Attribute Set:** R6 = {ScheduleID, DoctorID, ScheduleDate, StartTime, EndTime, AvailabilityStatus}

- **Functional Dependencies:**
    - FD6.1: ScheduleID → {DoctorID, ScheduleDate, StartTime, EndTime, AvailabilityStatus}

    - FD6.2: {DoctorID, ScheduleDate, StartTime} → {ScheduleID, EndTime, AvailabilityStatus} (since doctor shifts cannot overlap on the same date per BR24)
- **Candidate Keys:** CK1 = {ScheduleID}, CK2 = {DoctorID, ScheduleDate, StartTime}

Verification Steps:

1. **1NF:** Date, time, and status domains are atomic scalar types. ⇒ **1NF holds.**
2. **2NF:** No non-prime attribute depends on a proper subset of CK2 (e.g., DoctorID ↛ AvailabilityStatus). ⇒ **2NF holds.**
3. **3NF & BCNF:** Every determinant (ScheduleID and {DoctorID, ScheduleDate, StartTime}) is a superkey. ⇒ **BCNF holds.**

Proof for Relation: APPOINTMENT

- **Attribute Set:** R7 = {AppointmentID, PatientID, DoctorID, ScheduleID, BookedByUserID, FollowUpFromApptID, AppointmentDateTime, EstimatedDurationMinutes, AppointmentType, Status, Reason}

- **Functional Dependencies:**
    - FD7.1: AppointmentID → {PatientID, DoctorID, ScheduleID, BookedByUserID, FollowUpFromApptID, AppointmentDateTime, EstimatedDurationMinutes, AppointmentType, Status, Reason}

    - FD7.2: {DoctorID, AppointmentDateTime} → {AppointmentID, PatientID, ScheduleID, BookedByUserID, FollowUpFromApptID, EstimatedDurationMinutes, AppointmentType, Status, Reason} (applies strictly to Active Appointments where Status NOT IN ('Cancelled', 'No-show')

as per BR35 and BR36, preventing BCNF violations when preserving historical records of cancelled appointments)

- **Candidate Keys:** CK1 = {AppointmentID}, CK2 = {DoctorID, AppointmentDateTime}

Verification Steps:

1. **1NF:** All attributes hold atomic scalar values. ⇒ **1NF holds.**
2. **2NF:** No partial key dependency exists; doctor or patient personal metadata (name, contact) are factored out to their respective tables, preventing partial dependencies on subsets of CK2. ⇒ **2NF holds.**
3. **3NF:** No non-prime attribute determines another non-prime attribute. ⇒ **3NF holds.**
4. **BCNF:** Determinants are superkeys. ⇒ **BCNF holds.**

Proof for Relation: CONSULTATION_SESSION

- **Attribute Set:** R8 = {SessionID, AppointmentID, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingURL, DiagnosisNotes}

- **Functional Dependencies:**
  - FD8.1: SessionID → {AppointmentID, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingURL, DiagnosisNotes}

  - FD8.2: AppointmentID → {SessionID, StartTime, EndTime, ActualDurationMinutes, SessionType, MeetingURL, DiagnosisNotes} (Because every appointment yields at most one consultation session per BR38)
- **Candidate Keys:** CK1 = {SessionID}, CK2 = {AppointmentID}

Verification Steps:

1. **1NF:** Timestamps, URLs, and notes are atomic values. ⇒ **1NF holds.**
2. **2NF:** Candidate keys are singletons, precluding partial dependency. ⇒ **2NF holds.**
3. **3NF & BCNF:** FD8.1 and FD8.2 both originate from candidate keys (superkeys). The computed field ActualDurationMinutes is calculated strictly from (EndTime - StartTime) and constrained by domain rules. The removal of redundant foreign keys (DoctorID, PatientID) eliminates transitive dependencies, as these attributes are now implicitly available through the relationship chain (Session → Appointment → Patient/Doctor), ensuring the relation strictly adheres to BCNF. ⇒ **BCNF holds.**

Proof for Relation: MEDICAL_HISTORY

- **Attribute Set:** R9 = {MedicalHistoryID, SessionID, RecordDate, Diagnosis, Symptoms, ProgressNotes}

- **Functional Dependencies:**
  - FD9.1: MedicalHistoryID → {SessionID, RecordDate, Diagnosis, Symptoms, ProgressNotes}

- **Candidate Keys:** CK1 = {MedicalHistoryID}

- **Verification:** The only non-trivial functional dependency is `MedicalHistoryID → {SessionID, RecordDate, Diagnosis, Symptoms, ProgressNotes}`, and `MedicalHistoryID` is the candidate key of the relation. Therefore, the determinant `MedicalHistoryID` is a superkey. Since there are no non-key determinants that functionally determine other attributes, the relation satisfies BCNF. ⇒ **BCNF holds.**
-

Proof for Relation: PRESCRIPTION

- **Attribute Set:** R10 = {PrescriptionID, SessionID, DiagnosisICD, PrescriptionDate, Instructions}

- **Functional Dependencies:**
  - FD10.1: PrescriptionID → {SessionID, DiagnosisICD, PrescriptionDate, Instructions}

- **Candidate Keys:** CK1 = {PrescriptionID}

- **Verification:** The only non-trivial functional dependency is `PrescriptionID → {SessionID, DiagnosisICD, PrescriptionDate, Instructions}`, and `PrescriptionID` is the candidate key of the relation. Therefore, every determinant in the relation is a candidate key. Since there are no non-key determinants that functionally determine other attributes, the relation satisfies BCNF. ⇒ **BCNF holds.**

Proof for Relation: PRESCRIPTION_ITEM

- **Attribute Set:** R11 = {PrescriptionItemID, PrescriptionID, MedicationID, Dosage, Frequency, Duration, SpecialInstructions}

- **Functional Dependencies:**
  - FD11.1: PrescriptionItemID → {PrescriptionID, MedicationID, Dosage, Frequency, Duration, SpecialInstructions}

  - FD11.2: {PrescriptionID, MedicationID} → {PrescriptionItemID, Dosage, Frequency, Duration, SpecialInstructions}

- **Candidate Keys:** CK1 = {PrescriptionItemID}, CK2 = {PrescriptionID, MedicationID}

Verification Steps:

1. **1NF:** Each dosage instruction, frequency, and duration unit is atomic. ⇒ **1NF holds.**
2. **2NF:** The non-prime attributes (Dosage, Frequency, Duration, SpecialInstructions) depend upon the complete pair {PrescriptionID, MedicationID}. Medication descriptive names are stored in MEDICATION, ensuring no partial dependency (MedicationID ↛ MedicationName within this relation). ⇒ **2NF holds.**

3. **3NF & BCNF:** Both determinants are superkeys. ⇒ **BCNF holds.**

Proof for Relation: MEDICATION

- **Attribute Set:** R12 = {MedicationID, MedicationName, Description}
- **Functional Dependencies:**
  - FD12.1: MedicationID → {MedicationName, Description}
  - FD12.2: MedicationName → {MedicationID, Description}
- **Candidate Keys:** CK1 = {MedicationID}, CK2 = {MedicationName}
- **Verification:** Determinants are superkeys. No transitive dependencies exist. ⇒ **BCNF holds.**

3. DATA DICTIONARY (ADAPTED FROM ISO/IEC 11179)

This Data Dictionary standardizes the metadata definitions, representation formats, domain constraints, and relational roles for all tables in the Healthcare Clinic & Telemedicine Portal.

Table 3.1: USER_ACCOUNT

Description: Stores system authentication credentials, role definitions, and access statuses across the three authorized roles.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
UserID	VARCHAR(20)	No	None	**PK**. Unique alphanumeric user identifier.
Username	VARCHAR(50)	No	None	**Unique**. System login username.

PasswordHash	VARCHAR(255)	No	None	Cryptographic password hash (e.g., Argon2/Bcrypt).
Role	VARCHAR(10)	No	None	CHECK (Role IN ('ADMIN', 'DOCTOR', 'PATIENT')).
AccountStatus	VARCHAR(15)	No	'Active'	CHECK (AccountStatus IN ('Active', 'Locked', 'Suspended')).
CreatedAt	DATETIME	No	CURRENT_TIMESTAMP	Account provisioning audit timestamp.

Table 3.2: PATIENT

Description: Stores baseline demographic records and static medical baselines of registered patients.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
PatientID	VARCHAR(20)	No	None	**PK**. Unique patient master identifier.
UserID	VARCHAR(20)	No	None	**FK**, **Unique**. References

				USER_ACCOUNT(User ID).
FullName	VARCHAR(100)	No	None	Legal full name of patient.
DateOfBirth	DATE	No	None	Date of birth (must be ≤ CURRENT_DATE).
Gender	VARCHAR(10)	No	None	CHECK (Gender IN ('Male', 'Female', 'Other')).
Phone	VARCHAR(15)	No	None	Primary mobile phone number.
Email	VARCHAR(100)	Yes	NULL	Personal email address.
Address	VARCHAR(255)	Yes	NULL	Residential address.
BloodType	VARCHAR(5)	Yes	NULL	CHECK (BloodType IN ('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-')).
Allergies	TEXT	Yes	NULL	Baseline drug/food allergy history.
ChronicDiseases	TEXT	Yes	NULL	Pre-existing chronic diseases.

EmergencyContactName	VARCHAR(100)	No	None	Contact name for medical emergencies.
EmergencyContactPhone	VARCHAR(15)	No	None	Primary phone number of emergency contact.

Table 3.3: DOCTOR

Description: Superclass entity storing universal personal and professional doctor credentials.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
DoctorID	VARCHAR(20)	No	None	**PK**. Unique doctor identifier.
UserID	VARCHAR(20)	No	None	**FK**, **Unique**. References USER_ACCOUNT(UserID).
FullName	VARCHAR(100)	No	None	Full professional name of physician.
Phone	VARCHAR(15)	No	None	Official contact telephone number.
Email	VARCHAR(100)	No	None	**Unique**. Professional institutional email.
LicenseNumber	VARCHAR(50)	No	None	**Unique**. Government medical practice license code.

Table 3.4: GENERAL_PRACTITIONER

Description: Subclass entity designating General Practitioners (Disjoint and Total EER subtype).

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
DoctorID	VARCHAR(20)	No	None	**PK**, **FK**. References DOCTOR(DoctorID) on delete cascade.

Table 3.5: SPECIALTY

Description: Catalog of recognized clinical specialties.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
SpecialtyID	VARCHAR(20)	No	None	**PK**. Unique specialty identifier.
SpecialtyName	VARCHAR(100)	No	None	**Unique**. Official title of clinical specialty.
Description	TEXT	Yes	NULL	Detailed clinical scope description.

Table 3.6: SPECIALIST

Description: Subclass entity linking specialists to their assigned medical specialty.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning

DoctorID	VARCHAR(20)	No	None	**PK**, **FK**. References DOCTOR(DoctorID) on delete cascade.
SpecialtyID	VARCHAR(20)	No	None	**FK**. References SPECIALTY(SpecialtyID).

Table 3.7: DOCTOR_SCHEDULE

Description: Records scheduled clinical working periods and shifts for doctors.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
ScheduleID	VARCHAR(20)	No	None	**PK**. Unique shift schedule identifier.
DoctorID	VARCHAR(20)	No	None	**FK**. References DOCTOR(DoctorID).
ScheduleDate	DATE	No	None	Shift duty date.
StartTime	TIME	No	None	Shift start time.
EndTime	TIME	No	None	Shift end time (EndTime > StartTime).
AvailabilityStatus	VARCHAR(15)	No	'Available'	CHECK (AvailabilityStatus IN ('Available', 'Busy', 'On Leave')).

Table 3.8: APPOINTMENT

Description: Manages patient appointment reservations, follow-ups, and booking channels.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
AppointmentID	VARCHAR(20)	No	None	**PK**. Unique appointment reservation identifier.
PatientID	VARCHAR(20)	No	None	**FK**. References PATIENT(PatientID).
DoctorID	VARCHAR(20)	No	None	**FK**. References DOCTOR(DoctorID).
ScheduleID	VARCHAR(20)	No	None	**FK**. References DOCTOR_SCHEDULE(ScheduleID).
BookedByUserID	VARCHAR(20)	No	None	**FK**. References USER_ACCOUNT(UserID) (Patient, Doctor, Admin).
FollowUpFromApptID	VARCHAR(20)	Yes	NULL	**FK**. Recursive reference to preceding APPOINTMENT(AppointmentID). FollowUpFromApptID is NULL if the appointment is not a follow-up. If non-NULL, it must reference an existing valid preceding APPOINTMENT(AppointmentID).

AppointmentDateTime	DATETIME	No	None	Scheduled consultation date and time.
EstimatedDurationMinutes	INT	No	30	CHECK (EstimatedDurationMinutes IN (15, 30, 45, 60)).
AppointmentType	VARCHAR(15)	No	'In-person'	CHECK (AppointmentType IN ('In-person', 'Telemedicine')).
Status	VARCHAR(15)	No	'Scheduled'	CHECK (Status IN ('Scheduled', 'Checked-In', 'Completed', 'Cancelled', 'No-show')).
Reason	TEXT	Yes	NULL	Chief medical complaint or booking rationale.

Table 3.9: CONSULTATION_SESSION

Description: Records physical and telemedicine clinical encounters; enforces mandatory prior appointment.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
SessionID	VARCHAR(20)	No	None	**PK**. Unique clinical consultation session identifier.
AppointmentID	VARCHAR(20)	No	None	**FK**, **Unique**. References APPOINTMENT(AppointmentID).

StartTime	DATETIME	No	None	Actual timestamp when consultation commenced.
EndTime	DATETIME	No	None	Actual timestamp when consultation concluded (EndTime > StartTime).
ActualDurationMinutes	INT	No	None	CHECK (ActualDurationMinutes BETWEEN 5 AND 120).
SessionType	VARCHAR(15)	No	None	CHECK (SessionType IN ('In-person', 'Virtual')).
MeetingURL	VARCHAR(255)	Yes	NULL	Secure teleconference link (Telemedicine sessions).
DiagnosisNotes	TEXT	Yes	NULL	Attending physician's clinical notes.

Table 3.10: MEDICAL_HISTORY

Description: Archival longitudinal clinical diagnosis dossiers logged per consultation session.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
MedicalHistoryID	VARCHAR(20)	No	None	**PK**. Unique clinical diagnostic record code.

SessionID	VARCHAR(20)	No	None	**FK**. References CONSULTATION_SESSION(SessionID).
RecordDate	DATETIME	No	CURRENT_TIMESTAMP	Official clinical diagnosis date and time.
Diagnosis	VARCHAR(255)	No	None	Established diagnosis title/code.
Symptoms	TEXT	No	None	Observed clinical signs and symptoms.
ProgressNotes	TEXT	Yes	NULL	Physician's longitudinal assessment notes.

Table 3.11: MEDICATION

Description: Standardized pharmaceutical drug formulary.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
MedicationID	VARCHAR(20)	No	None	**PK**. Unique drug identifier code.
MedicationName	VARCHAR(100)	No	None	**Unique**. Generic or brand formulation title.

Description	TEXT	Yes	NULL	Pharmacological summary and contraindications.

Table 3.12: PRESCRIPTION

Description: Digital prescription headers issued during valid consultation encounters.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning
PrescriptionID	VARCHAR(20)	No	None	**PK**. Unique digital prescription identifier.
SessionID	VARCHAR(20)	No	None	**FK**. References CONSULTATION_SESSION(SessionID).
DiagnosisICD	VARCHAR(20)	Yes	NULL	ICD-10 diagnostic code.
PrescriptionDate	DATETIME	No	CURRENT_TIMESTAMP	Issuance timestamp.
Instructions	TEXT	Yes	NULL	Overall physician instructions to patient.

Table 3.13: PRESCRIPTION_ITEM

Description: Line-item medications, dosage parameters, and durations linked to a prescription.

Attribute Name	Data Type	Nullable	Default	Constraints & Business Meaning

PrescriptionItemID	VARCHAR(20)	No	None	**PK**. Unique line item identifier.
PrescriptionID	VARCHAR(20)	No	None	**FK**. References PRESCRIPTION(PrescriptionID).
MedicationID	VARCHAR(20)	No	None	**FK**. References MEDICATION(MedicationID).
Dosage	VARCHAR(50)	No	None	Prescribed strength (e.g., '500 mg').
Frequency	VARCHAR(50)	No	None	Administration schedule (e.g., 'Twice daily').
Duration	VARCHAR(50)	No	None	Course of treatment (e.g., '7 days').
SpecialInstructions	VARCHAR(255)	Yes	NULL	Intake guidelines (e.g., 'Take after meals').